PAS 4000:2026

Cargo handling facilities – Digitalization of logistic operations – Specification



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Contents

Foreword ... ii
0 Introduction ... iv
0.1 General ... iv
0.2 PAS 4000 role in Big Data analytics ... v

1 Scope ... 1

2 Normative references ... 2

3 Terms, definitions and abbreviated terms ... 3
3.1 Terms and definitions ... 3
3.2 Abbreviated terms ... 4

4 Semantic language of the dataset ... 5
4.1 General ... 5
4.2 Header ... 6
4.3 Subject ... 7
4.4 Concept ... 7
4.5 Observed property ... 7
4.6 Measurement essential ... 7
4.7 Payload ... 9

5 Data interchange ... 10
5.1 Hierarchical formats ... 10
5.2 Unique payload identifier (UPI) ... 12

6 Data model ... 13

Annexes

Annex A (informative)
Observed properties in the context of logistic operations ... 16

Annex B (informative)
Examples of messages in JSON, FLAT and UPI formats ... 18

Bibliography ... 21

List of figures

Figure 1 – Domain covered by PAS 4000 ... v
Figure 2 – Hierarchy of the elements used in a message ... 6
Figure 3 – Example of a hierarchical data structure in the case of a truck ... 11
Figure 4 – Structure of the data model ... 14
Figure 5 – Example of a message containing metadata ... 15
Figure 6 – Example of an observed property ... 15
Figure B.1 – Example of a message in JSON format ... 18
Figure B.2 – Example of a message in FLAT format ... 19
Figure B.3 – Example of a message in UPI format ... 20

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Foreword

This PAS was sponsored by TIC 4.0. Its development was facilitated by BSI Standards Limited and it was published under licence from The British Standards Institution. It came into effect on 28 February 2026.

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The provisions of this document are presented in roman (i.e. upright) type. Its requirements are expressed in sentences in which the principal auxiliary verb is "shall".

Commentary, explanation and general informative material is presented in smaller italic type, and does not constitute a normative element.

Requirements in this document are drafted in accordance with the *Rules for the structure and drafting of UK standards:2022*, subclause **G.1.1**, which states, "Requirements should be expressed using wording such as: 'When tested as described in Annex A, the product shall ...'". This means that only those products that are capable of passing the specified test will be deemed to conform to this document.

Where words have alternative spellings, the preferred spelling of the *Shorter Oxford English Dictionary* is used (e.g. "organization" rather than "organisation").

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In particular, attention is drawn to the following specific regulation:

- Regulation (EU) 2019/1239 of the European Parliament and of the Council of 20 June 2019 establishing a European Maritime Single Window environment and repealing Directive 2010/65/EU [1]

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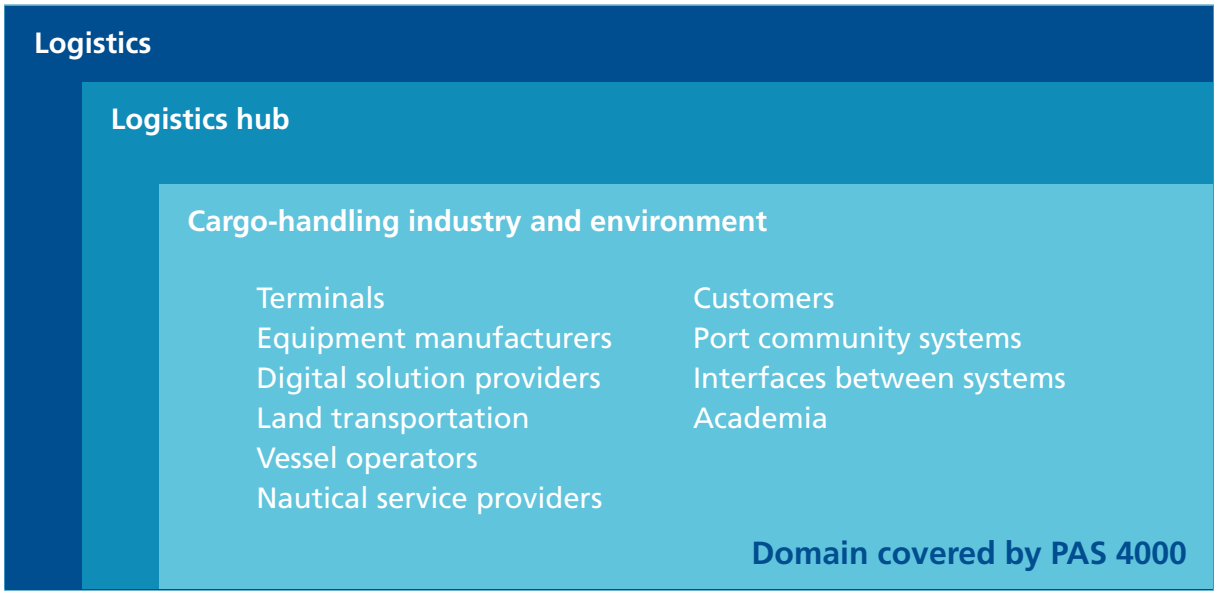
0 Introduction

0.1 General

PAS 4000 aims to describe and define a semantic model to represent, in a digital format, any reality in a specific, unambiguous and appropriate way, irrespective of the level of technology used.

PAS 4000 covers cargo handling facilities and their related software and hardware governing processes and operations. This includes hinterland operations and other logistics facilities (dry ports, warehouses, etc.). All the examples in PAS 4000 relate to container terminal handling facilities (see Figure 1).

Figure 1 – Domain covered by PAS 4000



The core function of the semantic model is to standardize data communications between key port facility equipment and its governing software and components. Connecting multiple physical objects with services present at a port terminal or alternative logistics facility and representing them in a digital environment is achieved through these common semantics and the semantic model. This way, data can be represented in an understandable format, including data and information elements that enable further data processing.

The semantic model provided in PAS 4000 can be applied to the following areas.

a) Machinery and equipment and all related information provided by them (e.g. sensors, software, condition, etc.).

b) Terminal processes and all the related information generated and collected by them.

c) Maintenance-related data operations and maintenance management systems (MMS).

d) Any other types of software that might use the data.

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Data generated by these systems' sensors and other types of attachments can be exchanged via any communications or telematics system.

NOTE 1 *At the time of publication, there is not a standardized language to handle or represent this data.*

PAS 4000 provides a semantic model only, and is primarily intended for ports and terminals. However, it can be applied to any sector to represent any reality when used with a dictionary that provides the required definitions for each element.

NOTE 2 *An example of a suitable dictionary is IMO's* Guidelines for setting up a maritime single window (FAL.5/Circ.42/Rev.4) *[2]. See Annex A, Note 1 for an example on how a dictionary might be applied in different use cases.*

PAS 4000 can be used with any version of a taxonomy, regardless of how many times that taxonomy has been updated.

PAS 4000 is fully compatible with any existing electronic data interchange (EDI), data list or data standard through the adoption of their vocabulary.

NOTE 3 *Some examples include the International Maritime Organization (IMO) Compendium on Facilitation and Electronic Business[1)], EDI-CEFACT[2)], SMDG standards[3)] and the Digital Container Shipping Association (DCSA) standards[4)].*

0.2 PAS 4000 role in Big Data analytics

The datasets generated by port facility equipment and software can be a valuable resource for management teams, as it provides information on the performance of equipment, infrastructure and organizational efficiency. PAS 4000 aims to facilitate this by providing requirements that can help organizations make this information:

a) comparable: enables benchmarking across systems and facilities;

b) interoperable: facilitates integration between different platforms and technologies;

c) solution-agnostic: no limitations on understanding by different systems and users;

d) specific: clearly identifies the process or equipment being referenced at a very high level;

e) unambiguous: provides clear and precise definitions to eliminate confusion for all users and systems; and

f) technology-level agnostic: usable in plain text (e-mails, documents, etc.) and embedded into protocols and electronic messages.

The use of these metrics for providing information can also achieve several key outcomes.

1) Enhanced connectivity between system modules and equipment from multiple suppliers, which reduces the need for problem-solving and can accelerate timeframes for individual projects.

2) Consistent and reliable information through standardized interpretation and definitions shared among all stakeholders.

3) Improved interoperability across vendors and suppliers.

4) Improved transparency, allowing better connectivity between terminals and other port stakeholders, including shipping lines and hinterland.

5) Elimination of ambiguity, providing clarity in the terminal operation environment.

6) Efficient project execution and terminal integration, allowing terminal operators and their suppliers to focus on operational improvements rather than technical interface and integration issues.

1) See https://www.imo.org/en/ourwork/facilitation/pages/imocompendium.aspx.
2) See https://unece.org/trade/uncefact/introducing-unedifact.
3) See https://smdg.org/.
4) See https://dcsa.org/standards/.

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1 Scope

PAS 4000 specifies requirements for the creation of sentences (see Clause **4**) to express any real-world action, status, event and process in a digital format for logistic operations within cargo-handling facilities.

The PAS aims to enable the digitalization of cargo-handling facilities and facilitate interoperability between different fully and semi-automated communication systems by providing common reference points for comparison, translation, harmonization of language and to move towards seamless data communication.

The PAS defines a set of elements of semantic language that can be used to cover:

a) all time instances:
   1) past/performed;
   2) present/actual;
   3) future/scheduled, planned, requested, proposed and estimated; and

b) all temporal data occurrences that:
   1) are instantaneous; or

      NOTE 1 *For example, "lights on status = true" or "lights power value = 50 W".*

   2) have a duration.

      NOTE 2 *For example, "lights on duration = 5 min" or "lights energy last hour value = 50 W/h".*

PAS 4000 covers the elements of header (see **4.2**), subject (see **4.3**), concept (see **4.4**), observed property (see **4.5**), measurement essentials (see **4.6**) and payload (see **4.7**) that, when combined, create a unique sentence which gives the payload a unique meaning that can be transmitted using any messaging protocols. The PAS provides examples.

PAS 4000 provides a semantic model only: definitions for the semantic elements are out of scope of this PAS. PAS 4000 is therefore intended to be used in conjunction with a dictionary in order to enable the PAS user to express almost any reality or any process in digital format.

PAS 4000 can be used with, but is not limited to, any telemetry application, electronic data interchange (EDI) or data list.

PAS 4000 does not cover IT/message protocols or any aspects of software or language coding in detail but does provide some examples.

PAS 4000 is for use by the following groups working within the cargo-handling and terminals industry, such as:

1) terminals and ports;
2) IT developers who might benefit from use of a common language to formulate requirements for IT and communication systems;
3) data scientists and those wanting to use a formal language to communicate the same meaning across different data formats and to effectively communicate locally or internationally;
4) individuals, such as senior or operations managers, who might wish to formulate queries to integrated data platforms in order to elicit information and data; and
5) equipment, technology and solution manufacturers and providers.

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This PAS might be of interest to regulators, port and other authorities, customs, shipping lines, cargo owners, freight-forwarders and academia.

PAS 4000 is intended to be used with any EDI, data list or data standards.

The PAS is intended for application within port terminals and related supply chains but can be used for other sectors.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes provisions, or limits the application, of this document[5)]. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

BS ISO 8601-1, *Date and time – Representations for information interchange – Basic rules*

5) Documents that are referred to solely in an informative manner are listed in the Bibliography.

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3 Terms, definitions and abbreviated terms

For the purposes of this document, the following terms and definitions apply.

3.1 Terms and definitions

3.1.1 array

fundamental data structure that stores a collection of elements, such as numbers, characters, or objects, in contiguous memory locations, allowing each element to be identified by a numerical value

3.1.2 array position

numerical value that indicates an element's place or order within an array

3.1.3 cargo

commercial or non-commercial item(s) to be transported

NOTE *The items(s) might or might not be loaded onto a carrier.*

3.1.4 carrier

entity responsible for carrying cargo

3.1.5 concept

element of semantic language which defines meta elements, actions, status and events performed by a subject or belonging to a subject

3.1.6 dictionary

set of precise and unambiguous definitions that provides precise meanings of the subjects, concepts and measurement essentials

3.1.7 hinterland

economic and geographical area, located landside, where a port exerts its influence in the form of sold goods and services, catchment area of cargo, infrastructure building and management

3.1.8 payload

actual data or content being transmitted in a communication, excluding any headers, metadata, or control information that are used for routing or processing the message

NOTE *Payloads are also commonly known as "values".*

3.1.9 physical location

geo-located physical location in the world or a specific location on a system or machine

NOTE *For example, an arm, a tyre, etc.*

3.1.10 terminal

enclosed area of a facility dedicated to the movement and storage of any type of cargo and the receival, loading and unloading of carriers

3.1.11 shipping container

container designed to transport cargo inside of it

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3.1.12 semantic model

conceptual framework represents the meanings and relationships of terms and concepts within a particular domain

NOTE The semantic model presented in PAS 4000 serves as a structured layer that abstracts and organizes physical data (such as database tables and columns) into logical entities, attributes, metadata, and relationships that reflect real-world business or application contexts.

3.1.13 subject

element of semantic language which defines who or what is being or doing the concept

3.1.14 taxonomy

hierarchy that defines how the information provided by different parties is to be structured and displayed and which establishes common semantics for the dataset and information structure

3.1.15 unique array identifier (UAI)

alphanumeric identifier that identifies every element of an array, no matter what array position the element occupies

3.1.16 unique payload identifier (UPI)

string that defines the payload's meaning in an unambiguous way

3.2 Abbreviated terms

For the purposes of this document, the following abbreviated terms apply.

EDI	electronic data interchange
ID	identifier
IoT	Internet Of Things
IP	internet protocol
GUID	globally unique identifier
HVAC	heating, ventilation, and air conditioning
JSON	JavaScript Object Notation
MMS	maintenance management system
MAC	medium access control
PLC	programmable logic controller
pomf	point of measurement – function
pomp	point of measurement – process
pomt	point of measurement – temporal
SI	International System of Units
UAI	unique array identifier
UPI	unique payload identifier
URL	uniform resource locator
XML	Extensible Markup Language

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4 Semantic language of the dataset

*COMMENTARY ON Clause **4***

PAS 4000 presents a common semantic language able to translate and consolidate the information that comes from multiple data sources. Due to this language, the semantic model enables the representation of any past, present or future reality, clearly defining who, when and what is executed. The semantic model provided in PAS 4000 is based on human-based language semantics, to address the difficulties that logistics actors face in their day-to-day communications.

4.1 General

The semantic language of the semantic model shall consist of six basic elements.

a) Header: unique identifier that identifies the message's origin, destination (if needed), reference time for the message and a unique ID (see **4.2**).

b) Subject: the entity that executes or embodies the concept (see **4.3**).

c) Concept: meta elements, actions, statuses and events performed by subjects or belonging to subjects. Concepts are tied to specific subjects (see **4.4**).

__NOTE 1__ An action is something that the subject performs, consumes or uses, for example, performing a movement, energy consumption or using motor oil or coolant. A concept defines what the subject is, what the subject does and/or what the subject consumes and always refers to a particular subject or subject-subsystem.

d) Observed property: the measured magnitude of a concept (see **4.5** and Annex A).

*__NOTE 2__ The actual payload of the magnitude is represented in the payload [see **4.1** f) and **4.7**]. Some examples of observed properties are: speed, acceleration, power, duration, volume, time, weight.*

e) Measurement essentials: all the metadata (information related to other data) that defines how the payloads are measured (see **4.6** and **5.2.2**).

f) Payload: set of attributes that result of a specific combination of time of measurement, subject, concept, observed property and point of measurement that reflect a unique reality (see **4.7**).

__NOTE 3__ Attributes are elements of semantic language that describe:

a) when an action is going to be performed;

b) how an action is going to be performed; and

c) where the action is going to be performed relative to the subject (input, output, internal input or output), both physically or as part of a process.

Attributes can be permanent (for example, a machine name or a container ID) or temporary (for example, a container's assigned port of discharge).

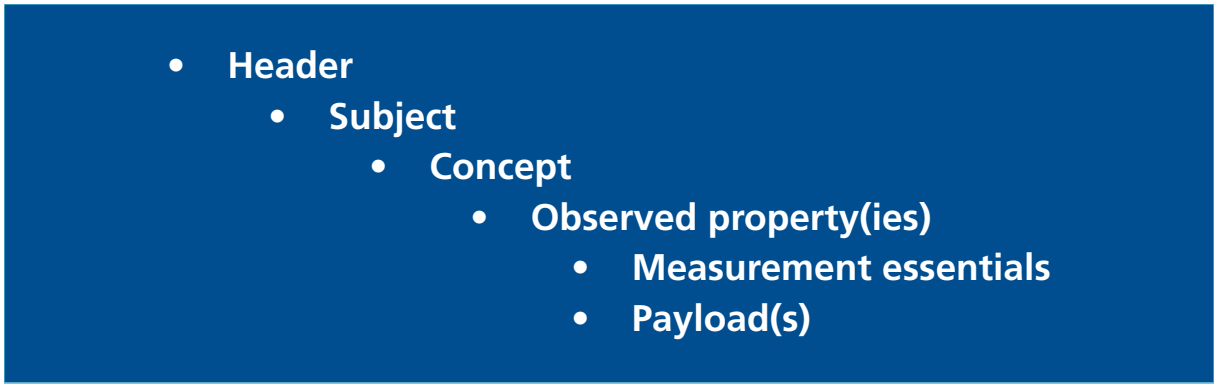
A message shall contain at minimum a subject and a concept.

The elements that constitute the message shall be unique.

The hierarchy of the elements used in a message shall follow the order provided in Figure 2.

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Figure 2 – Hierarchy of the elements used in a message



4.2 Header

The following header fields shall be used in the datasets.

a) "id": Message identifier; unique identifier reference for any message.

__NOTE 1__ The default format is the globally unique identifier (GUID) generator: a 128-bit number combined by e.g. MAC, timestamp and a random number combined by the sender's URL/IP.

b) "timestamp": Header message timestamp conforming to the format specified in BS ISO 8601-1, that includes the time zone indicator.

__NOTE 2__ Number code that defines the time when a message was generated (timestamp of the message creation), for example, 2024-06-21T12:35:08.000Z.

__NOTE 3__ Optional header fields that may also be used include, but are not limited to:

a) sender: header sender identifier; unique device identification that publishes the data, usually the subject's ID;

b) topic: defined as the information to be used for the publishing-subscribing technology (broker);

c) destination: unique device identification of the message's target;

d) creationtimestamp: Content creation time: the "creationtimestamp" is defined as the timestamp corresponding to when the message's content was created. This field is included if the message is issued at a different time than its content;

e) starttimestamp; endtimestamp: the "starttimestamp" and "endtimestamp" correspond to the start and end of the timeframe where the message's content is valid. This is necessary for aggregated values such as the average, maximum, counter, etc;

f) payloadtimestamp: default timestamp value of all payloads of the message;

g) protocol: the protocol language in which the message is written; and

h) version: version of the taxonomy that the message is using.

__NOTE 4__ The message header may be adapted to the requirements of other vendors or other languages.

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4.3 Subject

Subjects shall be either:

a) a physical entity such as a machine and any of its sub-systems, controllers and other objects (e.g. a vessel); or

b) an abstract entity such as a process, software program and any of its sub-systems (e.g. a vessel port visit).

NOTE 1 *If a relationship exists between subjects (one-to-one, many-to-one, one-to-many, or many-to-many), it is possible to nest the subjects in the taxonomy; this is known as "subject nesting". Subject nesting refers to organizing data in a hierarchical or nested structure. This approach is frequently employed in document-based databases and data formats like JSON, XML, among others.*

Subject nesting is defined as data containing an unlimited number of observations under a single key, which might itself be nested within a larger structure. These higher-level structures can consist of multiple observations, not necessarily in equal numbers.

In PAS 4000, physical objects and processes are, in practical terms, treated equally in terms of subjects, as long as the physical object or the process is a concept (state of being) or performs a concept.

A subject shall have an identifier (ID).

NOTE 2 *The ID is the unique device identification of the data subject. The ID can be composed of a manufacturer name plus a manufacturer ID, usually a serial number.*

The ID shall be unique and not be restricted to specific data pools (e.g. a terminal group, a manufacturer, etc.).

4.4 Concept

All subjects shall have at least one concept:

a) "static" status of a subject, e.g. "idle" referring to a machine;

b) "dynamic action" status of a subject, e.g. "movingtocollect" referring to a machine that is currently moving to collect a piece of cargo; or

c) "events".

4.5 Observed property

If an observed property is required to create a message, it shall define the magnitude of the concept.

NOTE 1 *Annex A provides some examples of observed properties that might be used in logistical operations.*

The elements of an observed property array shall be unique.

NOTE 2 *There can be many observed properties and magnitudes.*

4.6 Measurement essential

4.6.1 Measurement essential name

The name of a measurement essential shall be delimited by "#".

NOTE *For example, #pomf#, #pomt#, #sensorname#, #forecasttime#.*

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4.6.2 Point of measurement – function (pomf)

The point of measurement – function (pomf) shall be the point where the measurement is performed in relation to the subject (e.g. input and output).

If no information on the pomf is present, then the default value shall be assumed as "output".

Attributes of the pomf that shall be used are:

a) input: external order issued to a subject requiring one action or status to be "observed" and represented by a value. For the process' subject, it is the input of the process start; and

__NOTE 1__ An input is any input, instruction or signal sent via the cabin, remote control, autonomous control, etc. to a subject (e.g. machinery and equipment, processes, a gate, etc.). Examples of internal inputs are "IN"; "Command"; "Request" and "Demand".

__NOTE 2__ Internal orders can come from the programmable logic controller (PLC) or subsystems or any kind of internal control. Examples of internal inputs are "Request", "Command", "PLC request", "Internal Order" "Instance".

b) output: direct information from a sensor about the action or status to be observed. The output is a direct measurement confirming the observed property's value(s). For a process' subject, it is the output of the process end.

__NOTE 3__ Examples of outputs are "OUT", "Result" and "Final result".

4.6.3 Point of measurement – Temporal (pomt)

The pomt shall represent the relationship between the payload's measurement time (when it is measured) and the payload's time (when it is represented).

If no information on the pomf is present, then the default value shall be assumed as "actual".

The relevant attribute of the pomt shall be selected from the following.

a) Scheduled: indicates the intention to perform an action or status (a concept) in the far future in a regular way, for example, a timetable.

b) Proposal: indicates the intention of the owner of the action to perform an action/status (concept) in the future in a specific way an on a specific time.

c) Request: indicates the requirement to the owner of the action to do an action/status (concept) in the future in a specific way and on a specific time.

d) Planned: indicates the intention to do an action/status (concept) in the future in a specific way and on a specific time. Planned points of measurement shall be published by owners of the action (i.e. the one that controls and executes the process).

__NOTE__ For example, the port's pilot service owns the "carriervisit|pilot|firsttimepilotonboard" point of measurement.

e) Estimated: defined as the calculation or forecast about what and when in the future an action/status is expected to happen.

f) Actual: indicates that the timestamp of the payload (time when the payload was represented) and the timestamp of the moment where the measurement is taken are equal (i.e., the actual value).

g) Performed: indicates that measured or observed value of the concept observed was measured, an action or status happened in the past, but without knowledge of the actual time when it was performed.

h) Historic: indicates that measured or observed value of the concept observed was measured, an action or status that happened in the past and with exact knowledge of when it happened.

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4.6.4 Point of measurement – process (pomp)

Pomp shall be used only to describe the status of a process, with the following meaning: Process Started or Process Ended.

NOTE *In the case that a concept has started but not ended, the "pomp" is started. By default, all values are referred to "pomp = ended".*

4.6.5 Unit

All values shall include their respective measurement unit, when applicable.

NOTE *Units should be expressed in SI units where practicable. However, it is customary for non-SI units to be used in the maritime sector, so non-SI units may be used if necessary: for example, "knot" for wind or vessel speed, "nautical mile" for distance, "psi" for tyre or oil pressure, etc. When no unit is expressed in the "unit" field, the unit is assumed to be in SI unless stated otherwise.*

4.6.6 Generic metadata

All other metadata required to describe the value or how it is measured, that are not included in any of the categories described in **4.6.1** to **4.6.5**, shall be included as measurement essentials.

NOTE *Metadata may include elements such as identifiers, definitions, relationships to other subjects or concepts, provenance, ownership, descriptions, coverage, usage rights and application environment.*

4.7 Payload

The payload shall be derived by the result of a unique, specific combination of:

a) timestamp;

b) subject;

c) concept;

d) observed property; and

e) measurement essential(s).

NOTE *The combination of a subject with one or more concepts, observed properties and point of measurements produces a unique "meaning" of the value. The numerical value of the meaning can differ depending on the measurement unit used. Therefore, multiple representations of a value can exist depending on the unit or scale used (e.g. 36 km/h or 10 m/s).*

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5 Data interchange

5.1 Hierarchical formats

*COMMENTARY ON **5.1***

The data interchange format used should be the JSON format or any other hierarchical format.

JSON is the default file format of many protocols such as REST and MQTT, widely used to share data or publish IoT data. Other possible hierarchical structures, such as XML, may also be used.

Hierarchical formats are the most suitable formats for the representation of these semantics as they can express an array of subjects (for several nested subjects) or observed properties (for different combinations of timestamps, points of measurements, names, etc.). These arrays give high flexibility due to the hierarchical structure of the format.

In a hierarchical format, the terms "parent" and "child" refer to the relationship between elements where:

a) the parent is a node or item that stands at a higher level in the hierarchy and may contain or represent one or more subordinate elements called children; and

b) the child is a node that descends from a parent and may itself be a parent to other nodes further down the hierarchy.

Key characteristics of parents and children are:

1) each child has only one parent (except the top-most node, often called the "root," which has no parent);

2) a parent can have zero, one, or multiple children; and

3) hierarchies can have multiple levels (multi-level nesting) with parents, children, grandchildren, etc.

This structure is commonly visualized as a tree with nodes (parents) branching to other nodes (children).

Annex B provides examples of messages in JSON, flat and UPI formats.

In a hierarchical format, the message shall be built as follows:

a) Subject(s):
   1) represented as arrays; and
   2) has, at minimum, one child (concept or subject).

   ***NOTE 1** Subjects can be parents of other subjects and concepts, and children of other subjects.*

b) Concept(s):
   1) parent(s) of at least one observed property;
   2) minimum of one concept which is tied to a particular subject;
   3) not represented as arrays;
   4) does not have duplicated observed properties;
   5) does not have attached subjects; and
   6) does not have attached concepts.

c) Observed property(ies):
   1) has a concept as a parent;
   2) has at minimum one payload as a child; and

   ***NOTE 2** Observed property(ies) can have payloads and essential measurements (metadata) as children.*

   3) represented as arrays.

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d) Measurement essential(s):

   1) has, at minimum, one observed property as a parent; and

   2) does not have children; only metadata values.

e) Payload(s):

   1) has, at minimum, one observed property as a parent; and

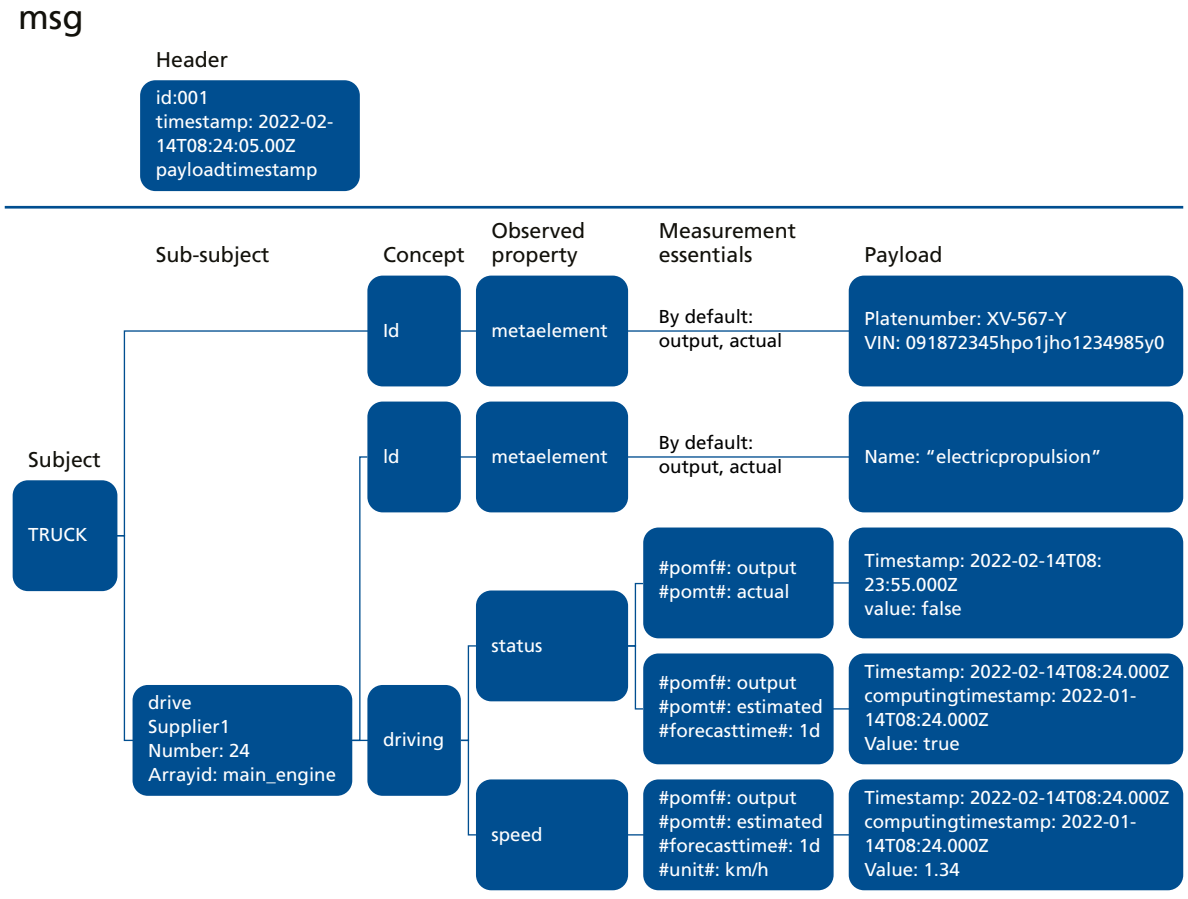
   2) does not have children; only metadata values.

f) Points of measurement: tied to observed properties.

NOTE 3 *The hierarchical structure allows the inclusion of an unlimited number of timestamps attached to concepts, subjects and nested subjects per the main subject. See Figure 3 for an real-life example of a hierarchical data structure, based on the operation of a truck.*

NOTE 4 *If a hierarchical data message has one unique timestamp for all observed properties, it can be included in the header of the message and omitted elsewhere.*

Figure 3 – Example of a hierarchical data structure in the case of a truck



NOTE 1 *See **4.6.2** for information on point of measurement: function.*

NOTE 2 *See Annex A for information on naming conventions and the meaning of the variables (e.g. Boolean, status. etc).*

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5.2 Unique payload identifier (UPI)

*COMMENTARY ON **5.2***

PAS 4000 provides payload names in hierarchy format.

A unique payload identifier (UPI) is a format to exchange data where only the "value key" is the UPI. UPIs enable data transmission as "value key" without any hierarchy. An advantage of UPI is that it provides all contexts required to understand the payload, and can be easier for humans to understand/express.

Annex B, Figure B.3 provides an example of a message in a UPI format.

Low-level implementations like PLCs or software code might use a "UPI" name. However, a flat format with UPI names is limited to including only one timestamp per message: see Annex B, Figure B.2 for an example of a message in a flat format.

5.2.1 Creating a UPI

A UPI shall be used if a payload needs to be expressed in a unique sentence. UPIs are expressed in the flat format.

A UPI shall consist of a string combination of the following elements, separated by a | symbol.

a) All subjects and nested subjects.

b) Unique array identifier (UAI), identified by a @, followed by the ID inside inverted comma(s), in the case of multiple arrays.

   ***NOTE 1** The purpose of a UAI is to identify the subject.*

c) Concept(s).

d) Observed property(ies).

e) All measurement essentials related to the payload names, while the metadata payload is located immediately after the # symbol that follows the metadata name.

f) The payload(s).

Each single, non-metadata payload shall be allocated to a single UPI.

***NOTE 2** Examples of UPIs:*

a) subject 1|@"unique array id"|concept|observedproperty|payloadname: payload

b) subject 1|@"unique array id"|nested subject 1|@"unique array id"|concept|observedproperty|#metadata1#metadatapayload1|#metadata2#metadatapayload2|payloadname: payload

c) Figure 3's example, expressed as a UPI:

truck|@|id|metaelement|platenumber: XV-567-Y

truck|@|id|metaelement|VIM: 091872345hpo1jho1234985y0

truck|@|drive|@|id|metaelement|name: electricpropulsion

truck|@|drive|@|driving|status|#pomf#output|#pomt#actual|value: false

truck|@|drive|@|driving|status|#pomf#output|#pomt#actual|timestamp: 2025-11-14T10:40:55|572Z

truck|@|drive|@|driving|status|#pomf#output|#pomt#estimated|#forecasttime#1day|computingtimestamp: 2025-11-14T10:40:55|572Z

truck|@|drive|@|driving|status|#pomf#output|#pomt#estimated|#forecasttime#1day|timestamp: 2025-11-15T10:40:55|572Z

truck|@|drive|@|driving|status|#pomf#output|#pomt#estimated|#forecasttime#1day|value: true

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truck|@|drive|@|driving|speed|#pomf#output|#pomt#estimated|#forecasttime#1day|#unit#km/h|computingtimestamp: 2025-11-14T10:40:55|573Z

truck|@|drive|@|driving|speed|#pomf#output|#pomt#estimated|#forecasttime#1day|#unit#km/h|timestamp: 2025-11-15T10:40:55|573Z

truck|@|drive|@|driving|speed|#pomf#output|#pomt#estimated|#forecasttime#1day|#unit#km/h|value: 1,34

5.2.2 Measurement essentials

Measurement essentials shall only be included if the specification of any temporal or spatial definition or metadata is required to make the measurement unique.

***NOTE** If no specification of any temporal or spatial definition or metadata is required, measurement essentials may be removed due to their permanent status.*

6 Data model

*COMMENTARY ON Clause **6***

The data model discussed in PAS 4000 has been defined based on the Resource Description Framework (RDF) *[3] using the subject → predicate → object schema.*

However, the data model's concept descriptions are not covered in this PAS. The data model is a technical implementation to represent these definitions in a digital format to work in hierarchical formats such as JSON, XML, HDF, etc. The data model as discussed in PAS 4000 can be applied to the following areas:

a) machinery and equipment and all related information provided by cargo-handling equipment (e.g. sensors, software, condition, etc.);

b) processes and all process-related information generated, collected and processed by other software and/or hardware; and

c) maintenance-related data operations and maintenance management systems (MMS).

Further extensions of the data model might cover other areas related to other agents of the logistics chain, port and customs authorities and others.

*As recommended in **5.1**, the preferred format is JSON. However, XML, MQTT, etc. can also be used if necessary.*

A data model shall be created to enable the digital formatting of the taxonomy and data set.

The data model used shall be based on the combination of the semantic model and the data model language elements provided in Clause **4**.

A schema shall also be created to define the details of the data model's content.

***NOTE 1** Examples of the details of the content include the validity of the format or the type of data (Boolean, integer, real etc.), what data is mandatory or what might be omitted, etc.*

***NOTE 2** If a JSON schema is used, a JSON validator should be used that verifies if the format is correct. From the JSON format, translation can be done to any other format (Java, YAMLK, XML etc.) or for human-readable format a flat format algorithm is also implemented.*

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The data model shall have three main components:

a) header;

b) asset description; and

c) measurement.

NOTE 3 This structure follows the semantics and ontology of the World Wide Web (subject: object). Figure 4 demonstrates the structure of this data model.

NOTE 4 Figure 5 illustrates an example of a message composed of:

a) three subjects (one subject plus two nested subjects);

b) three metadata per subject;

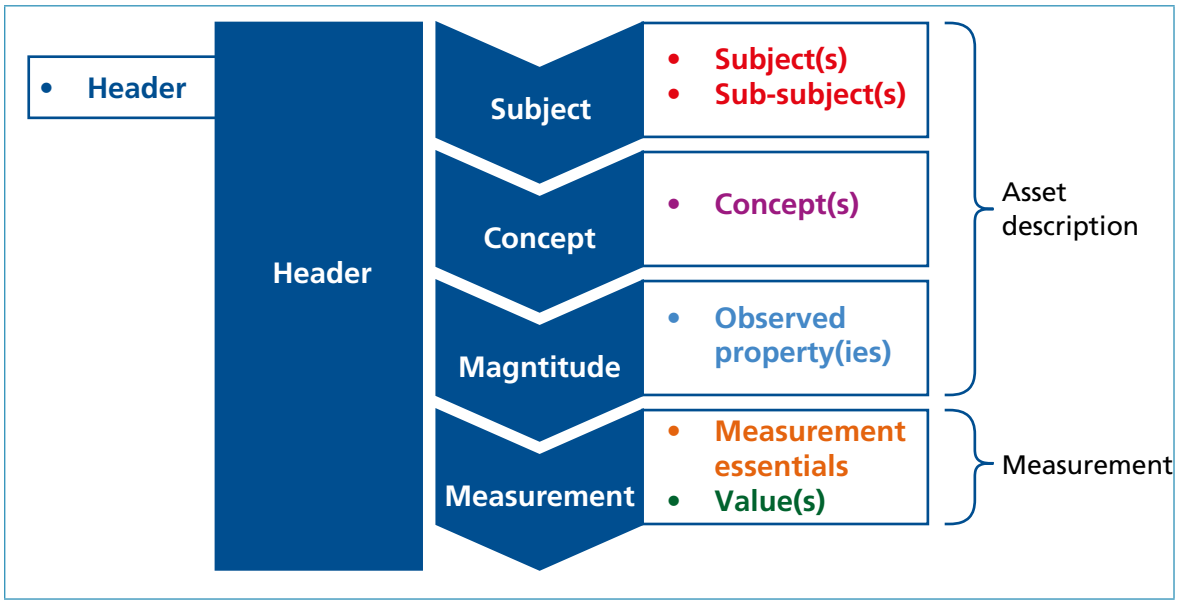
c) two different concepts that apply to several subjects; and

d) three observed properties that affect all the subjects.

Points of measurements and values have also been included in Figure 5 to provide clarification on their usage and position in the data model hierarchy if needed.

NOTE 5 Figure 6 illustrates an example of one observed property.

Figure 4 – Structure of the data model



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Figure 5 – Example of a message containing metadata



```
{
  "msg": {
    "header": {…
    }
  },
  "payload": {
    "subject1": [
      {
        "concept1": {
          "observedproperty1": [
            {
              "payload1": "value1",
              "payload2": "value2"
            }
          ]
        },
        "subject2": [
          {
            "concept1": {
              "observedproperty1": [
                {
                  "payload1": "value3",
                  "payload2": "value4"
                }
              ]
            },
            "concept2": {
              "observedproperty2": [
                {
                  "#measurementessential1#": "pomf1",
                  "#measurementessential2#": "pomt2",
                  "#measurementessential3#": "metadata5",
                  "#measurementessential4#": "metadata6",
                  "payload8": "value7",
                  "payload9": "value8",
                  "payload10": "value9"
                }
              ],
```

Figure 6 – Example of an observed property



```
"concept2": {
  "observedproperty2": [
    {
      "#measurementessential1#": "pomf1",
      "#measurementessential2#": "pomt2",
      "#measurementessential3#": "metadata5",
      "#measurementessential4#": "metadata6",
      "payload8": "value7",
      "payload9": "value8",
      "payload10": "value9"
    }
  ]
  "observedproperty3": [
    {
      "#measurementessential5#": "pomt3",
      "#measurementessential6#": "metadata7",
      "#measurementessential7#": "metadata8",
      "payload8": "value14",
      "payload9": "value15",
      "payload10": "value16"
    }
  ],
```

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Annex A (informative)
Observed properties in the context of logistic operations

Annex A illustrates some example definitions of common observed properties for logistic operations.

Annex A is not a dictionary and therefore does not provide a full list of observed properties. Users of PAS 4000 should therefore use the relevant dictionary of the sector or reality they wish to represent when applying this PAS.

NOTE 1 *Example of PAS 4000 applied to different use cases:*

a) *Use case example 1: ship finishing the arrival process: tos|@|carriervisit|@|arrival|@|end|time|#pomf#output|#pomt#actual|timestamp = 2025-06-13T12:58:52.294Z*

b) *Use case example 2: machine is reporting that it is ready to do the job designated for the present moment (output & actual by default): che|@|on|status|value = true*

c) *Use case example 3: one message with two planes reporting flight altitude, one at 4 340 feet altitude and another at 3 500:*

   plane|@1|flight|altitude|#sensorname#gps|value = 1323

   plane|@1|flight|altitude|#unit#feet|#sensorname#gps|value = 4340

   plane|@2|flight|altitude|#unit#feet|#sensorname#barometer|value 3500

*All names are given in UPI format (see **5.2**).*

The following is a non-exhaustive list of common observed properties and real-world examples.

a) Status: Boolean variable (i.e. “true” or “false”) representing a subject’s status in a determined timestamp.

   NOTE 1 *For example, when a truck is driving the status is “true” and when it is not driving it is “false”. UPI: truck|@|driving|status|value=true*

b) Duration (s): represents how long the status is, was or is expected be “true” just before the timestamp’s recording.

c) Counter (integer): a counter of the number of times the concept’s status has changed from “false” to “true” during a defined time period.

d) Distance: length travelled by a subject between two points during a period of time.

e) Speed (km/h) (m/min): derivative of the concept “distance” during a period of time.

f) Temperature (°C).

g) Pressure (Pa).

h) Voltage: difference in electric potential between two points.

i) Co-ordinate: latitude/longitude.

j) Metaelement: an observed property containing payloads to describe, explain, and provide context for the subject.

   NOTE 2 *For information on metadata, see **4.6.6**.*

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k) Humidity: measure of the amount of water vapour present in the air.

NOTE 3 This can be expressed in relative humidity.

l) Net weight: the weight of the subject itself, without the weight of any extra payload (e.g. container, packaging, etc.) it is carrying (the tare weight [see n)]).

m) Gross weight: the weight of the subject itself (the net weight [see l)]) plus the weight of any extra payload it is carrying (the tare weight [see n)]).

n) Tare weight: the weight of any extra payload the subject is carrying.

NOTE 4 The tare weight is the subject's gross weight [see m)] minus the subject's net weight [see l)].

This example list can be extended with other properties that are relevant to the combination of subject and concept.

Magnitudes can be expressed in units differing from the SI by indicating them in the field name #unit#.

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Annex B (informative)
Examples of messages in JSON, FLAT and UPI formats

Annex B provides examples of messages in several formats.

a) Figure B.1 provides an example of a message in JSON format.

b) Figure B.2 provides an example of a message in FLAT format.

c) Figure B.3 provides an example of a message in UPI format.

Figure B.1 – Example of a message in JSON format


```
{
  "msg": {
    "header": {…
    }
  },
  "payload": {
    "subject1": [
      {
        "concept1": {
          "observedproperty1": [
            {
              "payload1": "value1",
              "payload2": "value2"
            }
          ]
        },
        "subject2": [
          {
            "concept1": {
              "observedproperty1": [
                {
                  "payload1": "value3",
                  "payload2": "value4"
                }
              ]
            },
            "concept2": {
              "observedproperty2": [
                {
                  "#measurementessential1#": "pomf1",
                  "#measurementessential2#": "pomt2",
                  "#measurementessential3#": "metadata5",
                  "#measurementessential4#": "metadata6",
                  "payload8": "value7",
                  "payload9": "value8",
                  "payload10": "value9"
                }
              ],
              "observedproperty3": [
                {
                  "#measurementessential5#": "pomt3",
                  "#measurementessential6#": "metadata7",
                  "#measurementessential7#": "metadata8",
                  "payload8": "value14",
                  "payload9": "value15",
                  "payload10": "value16"
                }
              ],
```


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Figure B.2 – Example of a message in FLAT format

```
msg.header.id:
msg.header.idthread:
msg.header.idsequence:
msg.header.sender:
msg.header.timestamp: 2025-11-21T07:56:54.209Z
msg.header.topic:
msg.header.destination:
msg.header.creationtimestamp: 2025-11-21T07:56:54.209Z
msg.header.payloadtimestamp: None
msg.header.firsttimestamp: 2025-11-21T07:56:54.209Z
msg.header.lasttimestamp: 2025-11-21T07:56:54.209Z
msg.header.version:
msg.payload.subject1.0.concept1.observedproperty1.0.payload1: value1
msg.payload.subject1.0.concept1.observedproperty1.0.payload2: value2
msg.payload.subject1.0.subject2.0.concept1.observedproperty1.0.payload1: value3
msg.payload.subject1.0.subject2.0.concept1.observedproperty1.0.payload2: value4
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.#measurementessential1#: pomf1
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.#measurementessential2#: pomt2
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.#measurementessential3#: metadata5
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.#measurementessential4#: metadata6
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.payload8: value7
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.payload9: value8
msg.payload.subject1.0.subject2.0.concept2.observedproperty2.0.payload10: value9
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.#measurementessential5#: pomt3
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.#measurementessential6#: metadata7
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.#measurementessential7#: metadata8
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.payload8: value14
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.payload9: value15
msg.payload.subject1.0.subject2.0.concept2.observedproperty3.0.payload10: value16
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.#measurementessential8#: metadata9
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.#measurementessential9#: metadata10
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.#measurementessential10#: metadata11
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.payload11: value17
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.payload12: value18
msg.payload.subject1.0.subject2.0.concept2.observedproperty4.0.payload13: value19
```

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Figure B.3 – Example of a message in UPI format

```
subject1|@|concept1|observedproperty1|@|payload1: value1

subject1|@|concept1|observedproperty1|@|payload2: value2

subject1|@|subject2|@|concept1|observedproperty1|payload1: value3

subject1|@|subject2|@|concept1|observedproperty1|payload2: value4

subject1|@|subject2|@|concept2|observedproperty2|#measurementessential1#pomf1|#measurementessential2
#pomt2|#measurementessential3#metadata5|#measurementessential4#metadata6|payload8: value7

subject1|@|subject2|@|concept2|observedproperty2|#measurementessential1#pomf1|#measurementessential2
#pomt2|#measurementessential3#metadata5|#measurementessential4#metadata6||payload9: value8

subject1|@|subject2|@|concept2|observedproperty2|#measurementessential1#pomf1|#measurementessential2
#pomt2|#measurementessential3#metadata5|#measurementessential4#metadata6||payload10: value9

subject1|@|subject2|@|concept2|observedproperty3|#measurementessential5#pomt3|#measurementessential6
#metadata7|#measurementessential7#metadata8|payload8: value14

subject1|@|subject2|@|concept2|observedproperty3|#measurementessential5#pomt3|#measurementessential6
#metadata7|#measurementessential7#metadata8|payload9: value15

subject1|@|subject2|@|concept2|oservedproperty3|#measurementessential5#pomt3|#measurementessential6
#metadata7|#measurementessential7#metadata8|payload10: value16

subject1|@|subject2|@|concept2|observedproperty4|#measurementessential8#metadata9|
#measurementessential9#metadata10|#measurementessential10#metadata11|payload11: value17

subject1|@|subject2|@|concept2|observedproperty4|#measurementessential8#metadata9|
#measurementessential9#metadata10|#measurementessential10#metadata11|payload12: value18

subject1|@|subject2|@|concept2|observedproperty4|#measurementessential8#metadata9|
#measurementessential9#metadata10|#measurementessential10#metadata11|payload13: value19
```


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[3] WORLD WIDE WEB CONSORTIUM (W3C), RDF CORE WORKING GROUP. *Resource Description Framework* (RDF). W3C RDF Core Working Group: 2014.[8)]

Useful websites

Digital Container Shipping Association (DCSA) standards
https://dcsa.org/standards/

International Maritime Organization (IMO) Compendium on Facilitation and Electronic Business
https://www.imo.org/en/ourwork/facilitation/pages/imocompendium.aspx

SMDG standards
https://smdg.org/

TIC4.0 Atlassian repository of definitions
https://tic40.atlassian.net/wiki/spaces/TIC40Definitions/overview?homepageId=9699495

TIC4.0 standards
https://tic40.org/standards/

United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) Standards
https://unece.org/trade/uncefact/standards

United Nations rules for Electronic Data Interchange for Administration, Commerce and Transport (UN/EDIFACT)
https://unece.org/trade/uncefact/introducing-unedifact

6) Available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02019R1239-20250815.
7) Available at https://www.imo.org/en/ourwork/facilitation/pages/maritimesinglewindow-default.aspx.
8) Available at https://www.w3.org/RDF/.

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